



Agentic AI: a comprehensive survey of architectures, applications, and future directions

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Abstract

Agentic AI represents a transformative shift in artificial intelligence, but its rapid advancement has led to a fragmented understanding, often conflating modern neural systems with outdated symbolic models—a practice known as *conceptual retrofitting*. This survey cuts through this confusion by introducing a novel dual-paradigm framework that categorizes agentic systems into two distinct lineages: the symbolic/classical (relying on algorithmic planning and persistent state) and the neural/generative (leveraging stochastic generation and prompt-driven orchestration). Through a systematic PRISMA-based review of 90 studies (2018–2025), we provide a comprehensive analysis structured around this framework across three dimensions: (1) the theoretical foundations and architectural principles defining each paradigm; (2) domain-specific implementations in healthcare, finance, and robotics, demonstrating how application constraints dictate paradigm selection; and (3) paradigm-specific ethical and governance challenges, revealing divergent risks and mitigation strategies. Our analysis reveals that the choice of paradigm is strategic: symbolic systems dominate safety-critical domains (e.g., healthcare), while neural systems prevail in adaptive, data-rich environments (e.g., finance). Furthermore, we identify critical research gaps, including a significant deficit in governance models for symbolic systems and a pressing need for hybrid neuro-symbolic architectures. The findings culminate in a strategic roadmap arguing that the future of Agentic AI lies not in the dominance of one paradigm, but in their intentional integration to create systems that are both *adaptable* and *reliable*. This work provides the essential conceptual toolkit to guide future research, development, and policy toward robust and trustworthy hybrid intelligent systems.

Keywords Agentic AI · Artificial intelligence · Systematic review · Neural architectures · Symbolic AI · Multi-agent systems · AI governance · Neuro-symbolic AI

1 Introduction

The field of artificial intelligence (AI) is undergoing a paradigm shift from the development of passive, task-specific tools toward the engineering of autonomous systems that exhibit genuine agency. Modern Agentic AI systems (Wissuchek and Zschech 2025; Viswanathan et al. 2025) are defined by capabilities such as proactive planning, contextual memory, sophisticated tool use, and the ability to adapt their behavior based on environmental feedback. These systems operate not as mere solvers but as collaborative partners, capable of dynamically perceiving complex environments, reasoning about abstract goals, and orchestrating sequences of actions—either independently or as part of a sophisticated multi-agent ecosystem (Xie et al. 2024; Du et al. 2025).

To establish a precise conceptual foundation, we distinguish between the field's core concepts. An *AI Agent* (or a *single-agent system*) is a self-contained autonomous system designed to accomplish a goal. It operates primarily in isolation, though it may interact with tools and APIs. Its agency is defined by its *autonomy*, *proactivity*, and its ability to complete a task from start to finish independently.

For example, a single, powerful *LLM-based (large language model-based)* agent tasked with “Write a full project proposal for a new mobile app” would autonomously break down the task, conduct research, write the sections, and format the final document.

In contrast, *Agentic AI* is the broader field and architectural approach concerned with creating systems that exhibit agency. Crucially, this often involves the orchestration of *multi-agent systems (MAS)*, where multiple specialized agents work together, coordinating and communicating to solve problems that are too complex for a single agent.

For example, an Agentic AI system designed for the same task would employ a team of specialized agents: a *project manager agent* to break the goal into tasks, a *Researcher Agent* to gather market data, a *writer agent* to draft content, and a *quality assurance agent* to review the output. Their collaborative workflow is the embodiment of Agentic AI.

In summary, one can conceptualize an *AI Agent* as a single, sophisticated worker, while *Agentic AI* represents the principle of leveraging agency, frequently by architecting and managing an entire team of such workers.

This rapid evolution, however, has led to a fragmented and often anachronistic understanding of the field. A critical issue identified in prior reviews is *conceptual retrofitting*—the misapplication of classical symbolic frameworks (e.g., Belief–Desire–Intention (BDI) (Archibald et al. 2024), *perceive–plan–act–reflect (PPAR)* loops [Zeng et al. 2024; Erdogan et al. 2025]) to describe modern systems built on *large language models (LLMs)* (Plaat et al. 2025), which operate on fundamentally different principles of stochastic generation and prompt-driven orchestration. This practice obscures the true operational mechanics of LLM-based agents (Gabison and Xian 2025; Wang et al. 2024; Zhao et al. 2023; Chen et al. 2024) and creates a false sense of continuity between incompatible architectural paradigms.

Several recent reviews have explored aspects of Agentic AI, but most fall short of capturing its full scope or addressing the core challenge of conceptual retrofitting. As summarized in Table 1, existing surveys are often limited in scope, focusing on specific technical aspects, application domains, or high-level concepts without providing a unifying framework that acknowledges the fundamental paradigmatic schism.

Table 1 Summary of prior surveys on Agentic AI

References	Focus	Key contributions	Limitations
Plaata et al. (2025)	Agentic LLMs	Reasoning-acting-interacting taxonomy	Limited empirical validation
Schneider (2025)	GenAI → agentic shift	Conceptual framework for autonomy	No performance metrics
Acharya et al. (2025)	Foundational methods	Combined RL with cognitive architectures	Scalability not addressed
Gridach et al. (2025)	Scientific discovery	Tools for autonomous research workflows	No governance discussion
Hosseini and Seilani (2025)	Enterprise strategy	Agentic design for organizational alignment	Lack of technical depth
Sapkota et al. (2025)	Business operations	Systematic review methodology	Missing benchmark comparisons

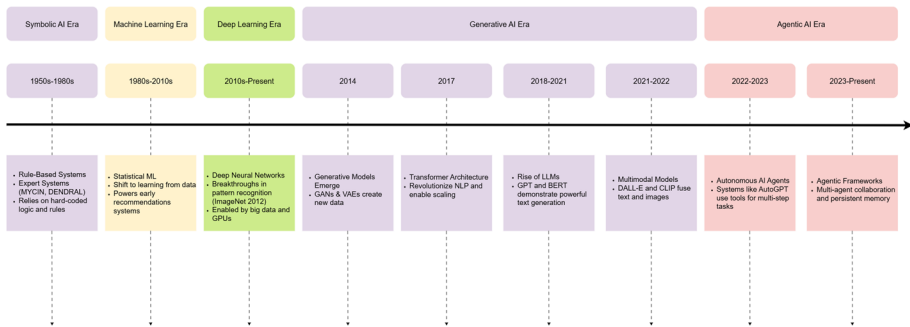


Fig. 1 Historical evolution of AI paradigms: this timeline charts the key breakthroughs and eras in AI, from early symbolic systems to the modern agentic era. It highlights the transformer architecture as the pivotal enabling technology for large language models (LLMs), which in turn powered the generative AI revolution and provided the substrate for contemporary agentic systems

Several recent reviews have explored aspects of Agentic AI, but most fall short of capturing its full scope or addressing core challenges. Table 1 summarizes their focus, contributions, and limitations.

This paper addresses these gaps by first establishing a clear historical context (Fig. 1), which delineates the evolution of AI through five distinct but overlapping eras.

The *Symbolic AI Era* (1950s–1980s) (Liang 2025) established the foundational ambition of artificial intelligence, grounded in logic and explicit human knowledge. This period was dominated by rule-based systems and expert systems such as MYCIN and DENDRAL (Swartout 1985), which operated on carefully hand-crafted symbolic rules. Intelligence was conceived as a top-down, deductive process, representing the purest form of the symbolic paradigm.

The *Machine learning (ML) Era* (1980s–2010s) (Thomas and Gupta 2020; Nithya et al. 2023; Trigka and Dritsas 2025) marked a pivotal shift away from hard-coded logic toward

systems that could learn from data. While still heavily dependent on human-engineered features, this period introduced statistical ML models such as Support Vector Machines and decision trees, which powered applications ranging from classification to recommendation. It was a transitional stage that moved the field away from pure symbolism but still lacked the automated feature learning that would define subsequent eras.

The arrival of the *Deep learning Era (2010s–present)* (Hatcher and Yu 2018; Alom et al. 2019; Dong et al. 2021; Khoei et al. 2023; Chhabra and Goyal 2023) was catalyzed by the confluence of increased compute power and large datasets. Deep neural networks, including convolutional and recurrent architectures, enabled systems to automatically learn hierarchical representations from raw data. This era revolutionized pattern recognition in vision, speech, and text, breaking longstanding barriers in perception. Yet, despite their power, these models largely functioned as sophisticated pattern classifiers rather than autonomous agents.

Out of this foundation emerged the *generative AI era (2014–present)* (Sakirin and Kusuma 2023; Anandhi 2025; Sengar et al. 2024; Surbakti 2025; Zhang et al. 2025), fueled by advances in generative modeling. Early breakthroughs such as generative adversarial networks were soon eclipsed by the introduction of the transformer architecture in 2017, which enabled the scaling of large language models (LLMs) such as GPT and BERT. These systems moved beyond perception to generation, producing coherent text, code, and media. In doing so, they provided the essential substrate—a powerful, general-purpose statistical reasoner—that made modern Agentic AI feasible.

Finally, the *Agentic AI era (2022–present)* represents the current frontier, where the generative capabilities of LLMs are harnessed for action and autonomy. This era is characterized by the rise of AI agents (Durante et al. 2024; Masterman et al. 2024; Piccialli et al. 2025) such as AutoGPT, which can pursue goals through planning and tool use. Increasingly, these agents evolve into multi-agent systems (Acharya et al. 2025; Viswanathan 2025; Plaat et al. 2025; Schneider 2025; Hosseini and Seilani 2025), exemplified by frameworks like CrewAI and AutoGen, where specialized roles and orchestrated collaboration enable teams of agents to tackle complex problems. In contrast to the algorithmic deliberation of the symbolic paradigm, this stage is defined by the neural paradigm, where agency emerges from the stochastic orchestration of generative models.

This chronological progression provides essential context but also reveals a critical conceptual schism. The Agentic AI era is not simply a linear descendant of symbolic AI but is instead built upon a completely different architectural foundation. To address this, we introduce a novel conceptual framework (Fig. 2) designed to prevent retrospective conflation by clearly distinguishing the symbolic and neural lineages of Agentic AI. This dual-axis taxonomy provides the unified lens necessary to rigorously analyze the field's theoretical underpinnings, architectural innovations, and practical deployments.

This paper addresses these gaps through four integrated contributions:

1. *A novel dual-paradigm taxonomy* We introduce and employ a new framework (Fig. 2) as our primary analytical tool, explicitly distinguishing symbolic and neural lineages to prevent conceptual retrofitting and enable accurate system classification.
2. *Architectural clarification* We demystify the operational principles of modern neural frameworks, explaining how they achieve agency through mechanisms like prompt chaining and conversation orchestration, rather than symbolic planning.

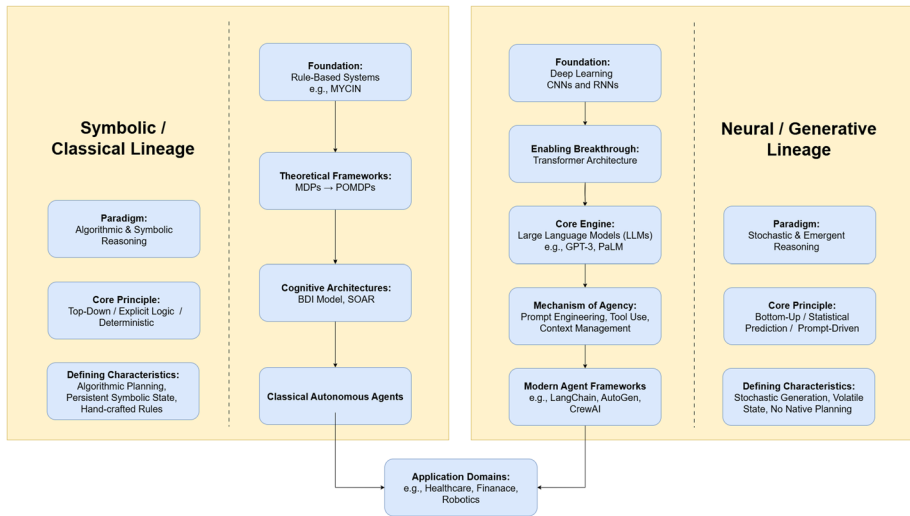


Fig. 2 Conceptual framework of Agentic AI’s dual lineages. This taxonomy resolves conceptual retrofitting by distinguishing the symbolic/classical lineage (left), defined by algorithmic planning and persistent state, from the neural/generative lineage (right), defined by stochastic generation and prompt-driven orchestration. While both paradigms target similar applications, their underlying mechanisms are fundamentally incompatible. This framework provides the analytical structure for this survey

3. *Empirical mapping* We conduct a systematic PRISMA-based literature review of 90 studies, categorizing them using our dual-paradigm framework to trace research trends and evaluate architectures by their appropriate standards.
4. *Governance anchoring* We embed ethical, accountability, and alignment challenges within each paradigm of our taxonomy to ensure that safety considerations are discussed in the correct technological context.

The paper is structured as follows: Sect. 2 presents our theoretical framework and dual-paradigm taxonomy. Section 3 details our systematic methodology. Section 4 presents our findings through a paradigm-aware analysis of the literature. Section 5 discusses the implications, limitations, and future directions revealed by our analysis. Section 6 concludes with a synthesis of key contributions.

2 Theoretical framework: a dual-paradigm taxonomy for Agentic AI

This paper introduces a novel dual-paradigm taxonomy (Fig. 2) to resolve conceptual retrofitting and provide a structured lens for analyzing the field. This framework serves as the primary analytical tool for this survey, categorizing agentic systems into two distinct lineages: the *symbolic/classical* and the *neural/generative*.

2.1 Core principles of autonomy and agency

The conceptual language for describing agency originated within the symbolic paradigm. The foundational constructs of *autonomy* and *agency* are essential for both lineages, though they are implemented in fundamentally different ways. Autonomy refers to a system's ability to operate independently, free from direct human intervention, whereas agency encapsulates the notion of goal-directed behavior that incorporates intention, contextual awareness, and decision-making capabilities (Patel et al. 2020; Kolt 2025). Agentic AI synthesizes these traits by initiating tasks, dynamically ranking goals, monitoring progress, and adjusting behavior through feedback loops (Trencsenyi et al. 2025).

These mechanisms parallel human executive functions such as planning, inhibition, and cognitive flexibility. They provide the high-level descriptive framework for intelligent behavior, which both symbolic and neural systems aim to achieve through divergent mechanisms.

2.2 The symbolic lineage: algorithmic decision-making

The symbolic lineage is characterized by explicit logic, algorithmic planning, and deterministic or probabilistic models. Its evolution provides the theoretical bedrock for pre-LLM autonomous systems.

2.2.1 Markov decision processes (MDPs)

MDPs provide the mathematical scaffolding for modeling environments with full state information (Lu et al. 2019, 2023), a hallmark of early symbolic and classical statistical AI. An MDP is defined by a tuple (S, A, P, R) , representing states, actions, transition probabilities, and rewards. These systems operate effectively in deterministic, rule-based domains but lack the capacity for robust reasoning under uncertainty, anchoring them firmly in the symbolic paradigm.

2.2.2 Partially observable MDPs (POMDPs)

POMDPs extend MDPs by introducing probabilistic *belief states* to handle environments where the agent has incomplete information (Rozek et al. 2024; Lu et al. 2024). This was a key advancement, allowing symbolic agents to infer hidden states through observation and enabling more adaptive behavior. However, as illustrated in Fig. 3, this is still a form of algorithmic state estimation. The significant computational overhead of belief tracking limits their scalability and real-world application (Frering et al. 2025; Gillen and Byl 2020), a fundamental constraint of the symbolic approach.

2.2.3 Cognitive architectures: BDI and SOAR

Cognitive architectures like belief-desire-intention (BDI) and State, operator, and result (SOAR) represent the pinnacle of the symbolic paradigm's attempt to engineer agency. SOAR is a specific, comprehensive cognitive architecture focused on modeling human-like problem-solving and learning. They explicitly model internal states and processes, as sum-

From Rule-Based scheduling to Belief-Based Inference

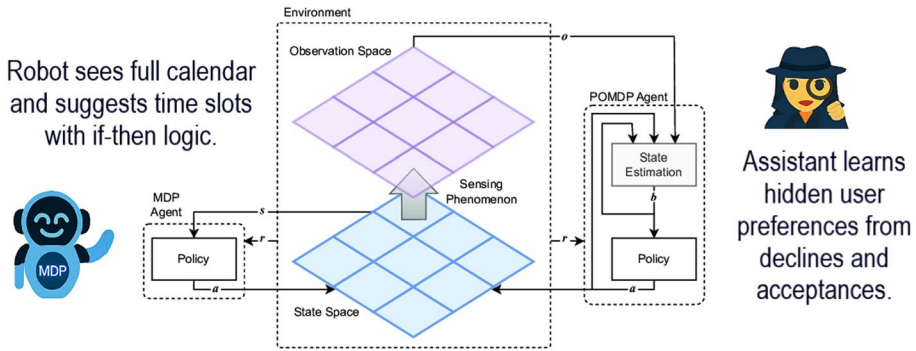


Fig. 3 Classical symbolic reasoning: comparison between a rule-based MDP scheduler (left) and a belief-based POMDP assistant (right). The MDP agent relies on explicit calendar states and deterministic policies, while the POMDP agent infers hidden user preferences from behavioral feedback. Both represent the symbolic paradigm’s approach to decision-making

Table 2 Mapping human cognitive functions to symbolic AI modules

Component	Human function	Symbolic AI parallel
Belief module	Working memory	Symbolic knowledge base/world model
Desire module	Motivation	Goal stack/utility function
Intention module	Executive control	Action policy/planner
Meta-cognition layer	Self-reflection, error monitoring	Monitor/replan loop

marized in Table 2. These systems directly implement a perceive-plan-act-reflect loop using symbolic representations, making them powerful but brittle and difficult to scale to complex, real-world environments. Their relationship to human cognitive functions is a direct, top-down mapping of symbolic logic.

2.3 The neural lineage: statistical learning and emergent reasoning

The neural lineage is built on a foundation of statistical learning from data, culminating in the generative capabilities of large language models (LLMs). Its progression is marked by a move away from explicit logic toward emergent, stochastic behavior.

2.3.1 Deep reinforcement learning (DRL)

Deep reinforcement learning (DRL) represents a critical transition. It scales learning to high-dimensional inputs (like images and text) using neural networks (Singh et al. 2025; Bodepudi et al. 2020). DRL agents learn policies directly from data, moving away from hand-crafted symbolic rules. methods such as proximal policy optimization (PPO) allow for fine-grained behavioral optimization (Kumar and Elumalai 2025; Yazid and Rachmawati 2023). PPO is a core DRL algorithm that enables stable and efficient learning of complex behaviors.

As shown in Fig. 4, advancements like meta-DRL introduced generalization across tasks, a precursor to the adaptability required for modern agency. DRL is a bridge, using neural networks to learn the policies that symbolic systems would have to be explicitly programmed with.

2.3.2 The LLM substrate and the paradigm shift

The emergence of large language models (LLMs) was not an evolution but a revolution that created the new neural paradigm. LLMs provided a powerful, general-purpose substrate for reasoning based on statistical prediction in a high-dimensional space of concepts. This enabled a fundamental architectural shift from designing cognitive agents to orchestrating generative pipelines.

Frameworks like LangChain (Mavroudis 2024), AutoGen (Wu et al. 2023), and CrewAI (Venkadesh et al. 2024; Duan and Wang 2024) do not implement symbolic PPAR loops

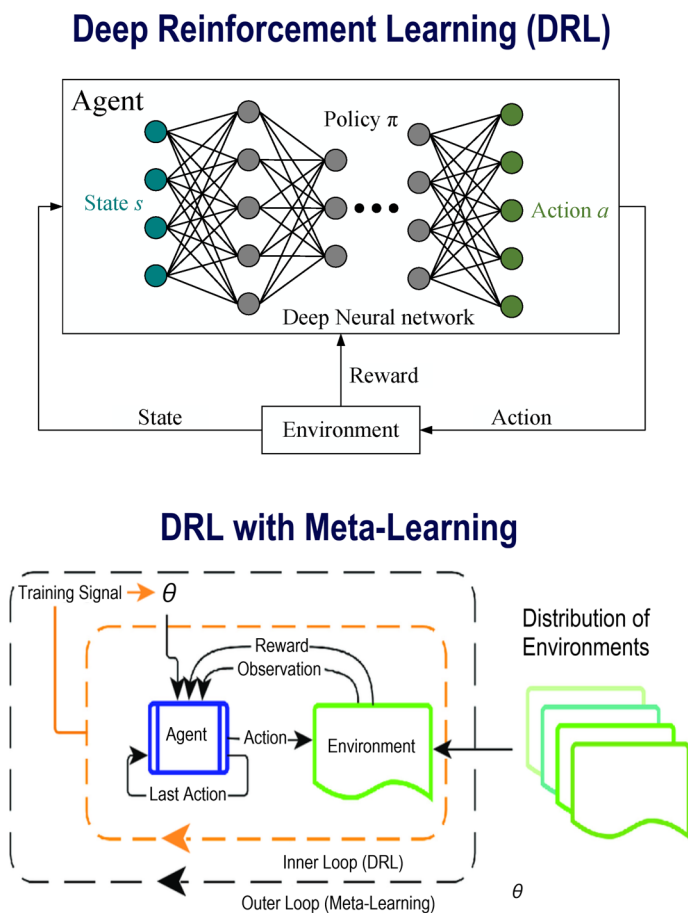


Fig. 4 The shift toward learned behavior: architectural contrast between vanilla DRL (single-task optimization) and meta-DRL (dual-loop generalization). The latter improves adaptability across tasks through meta-optimization loops, moving from explicit programming toward learned, emergent capabilities

or BDI architectures. They represent a new paradigm of *LLM orchestration*, where pre-trained models act as central executives that coordinate tasks through fundamentally different mechanisms, as detailed in Table 3.

This shift marks the definitive break from the symbolic tradition. Agency in the neural paradigm is an emergent property of prompt-driven orchestration, not a product of internal symbolic logic. The evolution of a personal assistant, depicted in Fig. 5, culminates in this new architecture.

2.4 Multi-agent orchestration: the pinnacle of the neural paradigm

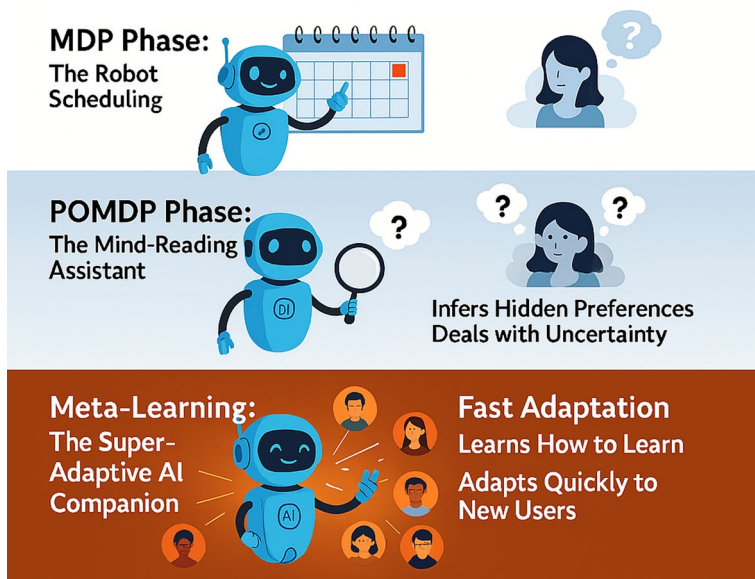
The most advanced manifestation of the neural paradigm is multi-agent orchestration. Frameworks like AutoGen (Wu et al. 2023) and LangGraph (Wang and Duan 2024) coordinate diverse, modular agents through structured communication protocols. As visualized in Fig. 6, an orchestrator (often an LLM itself) acts as a context manager and task router, assessing the overall goal and dynamically assigning specialized subtasks to other agents.

This architecture achieves scalability and complex problem-solving not through a single agent's cognitive complexity, but through the emergent intelligence of a well-orchestrated system. It is the culmination of the neural lineage, firmly establishing the new orthodoxy of LLM-driven pipelines and completing the paradigm shift from the symbolic AI tradition.

Table 3 Orchestration mechanisms of modern neural agentic frameworks

Framework	Primary mechanism	Functional paradigm (replaces traditional concept)
LangChain Mavroudis (2024), Johnson (2025), Gupta (2024), Topsakal and Akinci (2023) and Taulli and Deshmukh (2025)	Prompt chaining	Orchestrates linear sequences of LLM calls and API tools (replaces: symbolic planning)
AutoGen Wu et al. (2023) and Dibia (2025)	Multi-agent conversation	Facilitates structured dialogues between collaborative LLM agents (replaces: monolithic control)
CrewAI Venkadesh et al. (2024) and Duan and Wang (2024)	Role-based workflow	Assigns roles and goals to a team of agents, managing their interaction workflow (replaces: centralized scheduler)
Semantic Kernel Kothapalli (2024), Meyer (2024) and Costea (2025)	Plugin/function composition	Connects LLMs to pre-written code functions ("skills") for execution (replaces: integrated actuation)
LlamaIndex Gheorghiu (2024), Ramirez-Medina et al. (2025), Mozolevskiy and AISHikh (2024) and Braunschweiler et al. (2023)	Retrieval-augmented generation (RAG)	Efficiently retrieves and injects relevant information into the LLM's context (replaces: internal knowledge base)

From Basic Scheduling to Intuitive Personal Assistant: A Visual Journey



From Basic Scheduling to Intuitive Personal AI

Fig. 5 The journey from symbolic to neural agency: the evolution of a personal assistant from a deterministic rule-based (MDP) system, to an uncertainty-aware (POMDP) system, and finally to a modern LLM-orchestrated agent. This journey bridges the two paradigms, ending with a system that exhibits intelligent behavior through entirely different mechanisms

3 Methodology

A rigorous and transparent methodology is essential for constructing a comprehensive review that captures the dual paradigms of Agentic AI. This section outlines the systematic process used to identify, evaluate, and synthesize literature, with a specific focus on categorizing works according to the symbolic and neural lineages defined in our conceptual framework (Fig. 2). It follows established review protocols to ensure reproducibility while accounting for the field's rapid evolution.

3.1 Review design

This study adopts the *PRISMA 2020 framework* (preferred reporting items for systematic reviews and meta-analyses) (Page et al. 2021a, b), guiding all stages from search strategy to synthesis. The methodology is designed to capture and distinguish between the symbolic/classical and neural/generative lineages of Agentic AI research across computer science, cognitive psychology, robotics, and ethics.

Multi-Agent Workflow

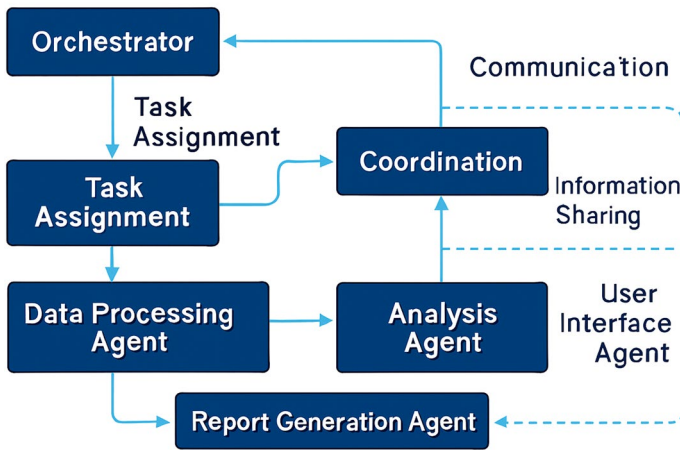


Fig. 6 The architecture of the neural paradigm: multi-agent orchestration in modern AI systems. This schematic illustrates the operational paradigm of neural systems. A central orchestrator (e.g., an LLM) manages a dynamic workflow of specialized agents through structured messaging and context management. Functionality emerges from prompt routing and API tool use, explicitly replacing the symbolic perceive-plan-act-reflect loop

Objectives This systematic review aims to provide a comprehensive analysis of Agentic AI systems through the following specific research objectives:

1. To identify, classify, and synthesize literature based on the dual architectural paradigms (symbolic vs. neural) of Agentic AI.
2. To examine the evolution of capabilities, applications, and performance metrics within and across each paradigm.
3. To analyze governance frameworks and ethical challenges, contextualizing them within their respective architectural paradigms.
4. To highlight paradigm-specific research gaps and propose informed future directions based on the synthesized evidence.

3.2 Data sources and search strategy

A multi-database search strategy was employed to identify literature across both historical symbolic and modern neural Agentic AI research. Sources included: IEEE Xplore, ACM Digital Library, arXiv, SpringerLink, ScienceDirect, and Google Scholar.

The search strategy employed a structured set of keyword clusters designed to comprehensively capture the core concepts associated with both architectural paradigms. To represent the *symbolic/classical* lineage, targeted terms included foundational concepts such as "Cognitive architectures", "BDI agent", "SOAR", "POMDP", "symbolic planning", and "multi-agent systems" (in its traditional sense). Conversely, the *neural/generative* paradigm was captured through terms reflecting its contemporary emergence, such as "LLM agent", "AI orchestration", "prompt chaining", "tool-augmented LLM", "multi-agent conversa-

tion", and specific framework names including "AutoGen" and "LangChain." Finally, a set of *General* terms—"Agentic AI," "autonomous agent," and "goal-directed AI"—was used to ensure broad coverage and to capture literature that might bridge or transcend the paradigmatic divide. Boolean operators were structured to optimize breadth and relevance (e.g., ("autonomous agent" OR "Agentic AI") AND ("large language model" OR "orchestration" OR "cognitive architecture")).

The search scope was interdisciplinary, targeting relevant fields from computer science to ethics. To capture the most current advancements in the rapidly evolving neural paradigm, the search included pre-print servers like arXiv, with these records being manually assessed for quality and relevance.

3.3 Inclusion and exclusion criteria

To ensure the review's methodological integrity and thematic relevance, predefined inclusion and exclusion parameters were applied during the screening process. These criteria were designed to capture high-quality literature from both paradigms of Agentic AI.

Inclusion criteria The literature search employed the following inclusion criteria to identify publications that contribute directly to the core themes of Agentic AI architectures and applications. Specifically, we included peer-reviewed journal articles, conference proceedings, and formally published technical reports from recognized institutions. To capture the most recent advancements in the rapidly evolving neural paradigm, we also incorporated high-impact pre-prints from arXiv, which were manually screened for methodological rigor and citation impact, with a focus on those presenting novel architectures or frameworks. The scope of included work encompassed studies involving the design, implementation, or evaluation of autonomous agents, spanning both classical symbolic systems and modern LLM-orchestrated frameworks. All selected publications were required to be in English and published within the temporal window of January 2018 to March 2025.

Exclusion criteria To ensure a focused and methodologically rigorous review, studies were excluded according to the following criteria. Non-English language publications were omitted. We also excluded non-peer-reviewed or informal sources such as opinion pieces, editorials, blog posts, and unverified online content. Furthermore, studies focused exclusively on generative AI (e.g., for image generation or text completion) without incorporating agentic features like goal-directedness, tool use, or multi-step autonomy were deemed out of scope. Finally, duplicate records retrieved from multiple databases were identified and removed to prevent redundancy in the analysis.

These criteria ensured the retention of conceptually aligned and methodologically sound studies from both paradigms, preserving the review's comprehensive scope. A summary is provided in Table 4.

3.4 Screening and selection process

The screening protocol adhered to the PRISMA 2020 guidelines to ensure methodological transparency and reproducibility. Records were compiled from selected databases, yielding an initial pool of 165 items (157 from databases, 8 from supplemental sources).

Table 4 Inclusion and exclusion criteria for literature selection

Category	Criteria
Inclusion	• Peer-reviewed journal and conference papers • Technical reports from reputable institutions • Studies on autonomous agents from both symbolic and neural paradigms • Applications across various domains demonstrating agentic capabilities • Published in English between 2018 and 2025
Exclusion	• Non-English publications • Blogs, opinion pieces, or informal content • Studies focused solely on generative AI without agentic autonomy • Duplicate records across multiple databases

Following deduplication, 120 unique records remained. Title and abstract screening excluded 42 studies due to irrelevance or insufficient focus on Agentic AI. Full-text assessment confirmed 78 articles met all inclusion criteria.

In alignment with PRISMA's guidance for systematic reviews that require foundational context, a supplemental phase was conducted (Page et al. 2021a). During thematic synthesis, 12 seminal theoretical papers from the symbolic paradigm [e.g., foundational works on MDPs by Kaelbling et al. (1998) and cognitive architectures by Laird (2022)] were incorporated. These papers were essential for providing complete historical context for the taxonomic framework and understanding the symbolic lineage, though they were analyzed separately from contemporary neural paradigm research. This resulted in a final corpus of 90 publications for contextual and theoretical grounding, with 78 studies forming the core for analysis of contemporary trends.

The process is illustrated in Fig. 7, which clearly distinguishes the primary systematic search from the supplemental inclusion of foundational context.

3.5 Data analysis

The 78 studies forming the core of the review underwent thematic synthesis following the methodology described by Thomas and Harden (2008), with analysis specifically structured around the dual-paradigm framework.

The synthesis was conducted following the framework by Thomas and Harden (2008) and was structured around the dual-paradigm taxonomy. The coding process was primarily *deductive*, using the pre-defined categories of the Symbolic and Neural paradigms as initial codes. Within these paradigms, *inductive* coding was applied to identify emergent themes related to architectures, applications, and challenges.

To ensure reliability, the coding was performed by two independent coders. An inter-coder reliability (ICR) check was conducted on a 15% subset of the papers, resulting in a Cohen's Kappa coefficient of 0.82, indicating substantial agreement. Discrepancies were resolved through discussion until consensus was reached. The final coding scheme was applied to the entire corpus using NVivo (Van et al. 2025) to facilitate analysis and manage the thematic structure.

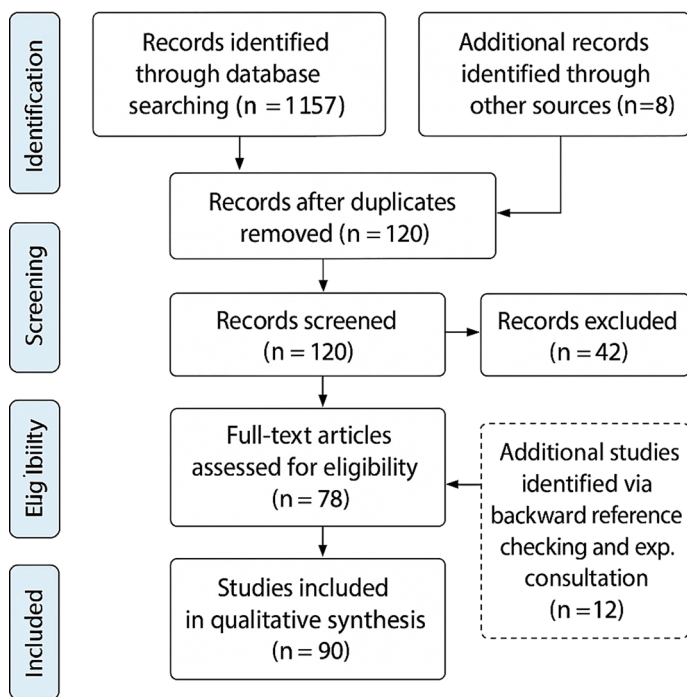


Fig. 7 PRISMA 2020 flow diagram. Records were identified from databases ($n=157$) and supplemental sources ($n=8$). After deduplication ($n=120$) and title/abstract screening ($n=42$ excluded), full-text review confirmed 78 eligible studies. A supplemental phase added 12 seminal theoretical papers for contextual framing of the symbolic paradigm (shown in dashed box), yielding a final corpus of 90 publications for the review

3.5.1 Key analytical techniques

Our analysis employed a multi-faceted methodological approach to systematically investigate the body of research. The initial phase involved *paradigm classification*, whereby each study was categorized according to its primary architectural paradigm—either symbolic/classical or neural/generative—based on the core operational mechanisms defined in our conceptual framework. Following this classification, we conducted a detailed *framework mapping* within each paradigm to group studies by their specific architectural approaches, including orchestration models (e.g., AutoGen, CrewAI), memory structures, and learning mechanisms. Building on this organized foundation, a *cross-paradigm comparison* was performed to identify fundamental differences in implementation, performance, and limitations between the two overarching paradigms. In parallel, we performed *domain clustering* to group applications by sector—such as healthcare, finance, robotics, and scientific discovery—which enabled the identification of performance patterns and deployment strategies both within and across paradigms. Finally, an *ethical coding* procedure was applied, using a structured lexicon to tag recurring themes related to governance, safety, transparency, and bias, with particular attention paid to how these ethical challenges manifest differently within each paradigm.

Qualitative coding was supported by tools such as NVivo (Van et al. 2025), which enabled hierarchical theme identification and cross-paradigm analysis. Quantitative results were tabulated and compared within and across domains and paradigms to synthesize technical and operational insights.

This paradigm-informed approach ensured a nuanced understanding of the current landscape of Agentic AI research, supporting both theoretical grounding and real-world applicability while maintaining the analytical rigor required for this review.

3.6 Limitations

Limitations While this review provides a comprehensive synthesis of Agentic AI research, several limitations must be acknowledged. First, the inherent *temporal and scope dynamics* of the field, particularly within the rapidly evolving neural paradigm, present a challenge; although our search extended to early 2025, some very recent developments may not be captured, a risk mitigated but not fully eliminated by the inclusion of pre-prints. Furthermore, our methodological approach required a *contextual reference expansion* through the supplemental inclusion of 12 seminal symbolic papers to ensure a robust theoretical framing of the classical lineage. We emphasize that these papers, analyzed separately from contemporary research, were used strictly for contextual and historical background and represent a deviation from a purely systematic retrieval process.

Additional constraints arose from the nature of the subject matter itself. *Transparency constraints* were encountered as many state-of-the-art neural agentic systems operate as proprietary solutions with limited public documentation, meaning architectural details and performance metrics were sometimes incomplete or inferred from secondary sources. *Methodological heterogeneity* across the reviewed studies, with their varied evaluation metrics, also limited our ability to perform direct cross-study benchmarking, particularly between paradigms that employ fundamentally different performance measures. Finally, despite implementing rigorous classification criteria, the *paradigm classification challenge* of assigning hybrid or transitional architectures to a single paradigm may, in some cases, involve necessary simplification.

These limitations collectively highlight the challenges of conducting systematic reviews in a nascent and fast-paced field with multiple co-existing paradigms. Our two-phase approach—a systematic review of contemporary research supplemented by a narrative inclusion of foundational symbolic context—was designed to balance methodological rigor with comprehensiveness while respecting the fundamental distinctions between these architectural paradigms.

4 Findings: a paradigm-aware analysis of the Agentic AI landscape

The results of our systematic review, synthesized through the lens of our dual-paradigm taxonomy, are presented in Fig. 8 and Table 5. This paradigm-aware analysis reveals several key patterns that were previously obscured in the literature.

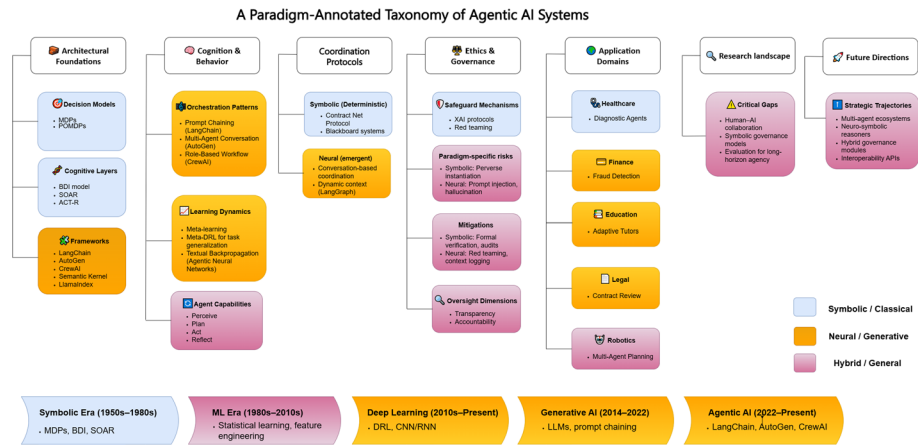


Fig. 8 A paradigm-annotated taxonomy of Agentic AI systems. This expanded framework organizes the Agentic AI landscape into seven modular domains, each color-coded by architectural paradigm: symbolic/classical (blue), neural/generative (orange), and hybrid/general (purple). A bottom timeline anchors these paradigms across five historical eras—from symbolic logic to agentic ecosystems—highlighting the evolution of cognition, coordination, and governance. The taxonomy now includes orchestration patterns (e.g., prompt chaining, multi-agent dialogue), coordination protocols (symbolic vs. emergent), and paradigm-specific risks with tailored mitigation strategies. Application domains are annotated by their dominant lineage, illustrating the trade-offs between symbolic rigor and neural adaptability. This visualization offers a strategic roadmap for designing, evaluating, and deploying agentic systems across diverse contexts

4.1 Paradigm-based classification of the literature

Our analysis of the complete corpus reveals three significant patterns:

1. *Paradigm specialization by domain* High-stakes, regulated domains like Healthcare and Legal Tech show a strong preference for symbolic or highly constrained neural architectures (e.g., Singh et al. 2024; Magesh et al. 2024), while dynamic domains like Finance leverage neural orchestration for complex analysis (e.g., Chandrashekar et al. 2025).
2. *The governance divide* Research in ethics and governance is overwhelmingly focused on the novel challenges of the neural paradigm (e.g., Gabison and Xian 2025; Raza et al. 2025), revealing a significant gap in modernized governance frameworks for purely symbolic systems.
3. *Temporal paradigm shift* The data shows a clear transition: symbolic and hybrid cognitive architectures dominated early research (2018–2021), while neural Orchestration Frameworks have overwhelmingly dominated post-2022, following the rise of LLMs.

4.2 Foundational studies: the roots of two lineages

The theoretical bedrock of Agentic AI is found in two distinct lineages, each with its own foundational breakthroughs. Landmark studies have shaped the conceptual and architectural foundations of both paradigms, spanning strategic reasoning, cognitive models, and align-

Table 5 Paradigm-based taxonomy of Agentic AI literature (2018–2025)

Category	Paradigm	Key papers	Year	Focus area	Key contributions
Foundational theories	Hybrid	Plaat et al. (2025), Schneider (2025), Acharya et al. (2025), Gridach et al. (2025), Hosseini and Seilani (2025) and Sapkota et al. (2025)	2025	Autonomy frameworks	Theoretical foundations bridging symbolic and neural concepts of agency
Architectural frameworks	Neural	Wu et al. (2023), Venkadesh et al. (2024), Duan and Wang (2024), Kothapalli (2024), Meyer (2024), Costea (2025), Gheorghiu (2024), Mavroudis (2024), Johnson (2025), Gupta (2024), Topsakal and Akinci (2023), Taulli and Deshmukh (2025) and Dibia (2025)	2023–2025	System design	Neural-based multi-agent orchestration, tool integration, and workflow management
Healthcare applications	Symbolic / hybrid	Singh et al. (2024), Huh et al. (2023), Basit et al. (2024) and Cárdenas et al. (2024)	2023–2024	Medical AI	Clinical decision support using deterministic and constrained neural systems for safety
Robotics and automation	Hybrid	Bai et al. (2025), Zhang et al. (2024), Annanth et al. (2021), Singireddy and Daim (2018) and Patel et al. (2020)	2018–2025	Autonomous systems	Combines symbolic planners (POMDPs) for safety with neural components for adaptability
Financial systems	Neural	Roychowdhury et al. (2023), Yang et al. (2023), Chandrashekar et al. (2025), Konstantinidis et al. (2024) and Al-E'mari et al. (2025)	2023–2025	FinTech	Neural agents for fraud detection, algorithmic trading, and risk assessment
Education technology	Neural	Suh (2025) and Fisher et al. (2020)	2020–2025	EdTech	Neural-based adaptive learning systems and intelligent tutoring
Legal and compliance	Neural (RAG)	Magesh et al. (2024)	2024	Legal tech	Neural agents heavily constrained by symbolic retrieval (RAG) for accuracy

Table 5 (continued)

Category	Paradigm	Key papers	Year	Focus area	Key contributions
Ethics and governance	Neural	Gabison and Xian (2025), Raza et al. (2025), Syros et al. (2025), Feng et al. (2025), Tallam (2025), Clatterbuck et al. (2024), Chan et al. (2023), Tóth et al. (2022), Baron (2025), Chan et al. (2024), Papagni et al. (2023), Singh and Ngu (2025), Yadav (2024), Hellrigel-Holderbaum and Dung (2025), Zhang et al. (2025), Kasirzadeh and Gabriel (2025), Tiwari (2025), Costa and Aparicio (2025), Lakkamraju (2025), Zou et al. (2025), Borghoff et al. (2025), Tennant et al. (2023) and Rossi and Mattei (2019)	2019–2025	AI safety	Frameworks addressing neural-specific challenges (alignment, bias, opacity)
Evaluation and benchmarking	Neural	Liu et al. (2023), Reuel et al. (2024), Yehudai et al. (2025), Zhuge et al. (2024) and Moshkovich et al. (2025)	2023–2025	Performance metrics	Benchmarks focused on neural agent capabilities (reasoning, tool use)
Emerging technologies	Hybrid	Sultanow et al. (2025), Yang et al. (2025), Agashe et al. (2014), Huang et al. (2025), Bachrach et al. (2020), Radanliev (2025), Ma et al. (2025), Schmidt and Loidolt (2023), Koutra et al. (2025), Lu et al. (2024), Cervantes et al. (2020), Nayak (2025), Karim et al. (2025), Thompson (2024), Bovo et al. (2025), Samuel et al. (2022), Wu et al. (2025), Zhuang et al. (2025), Jeong (2025), Li and Xie (2025), Summers et al. (2023), Romero et al. (2023), Shapiro et al. (2023) and Nong (2025); Li et al. (2025)	2020–2025	Innovation frontiers	Research into neuro-symbolic integration, quantum AI, and human-AI collaboration

ment. A summary of these contributions is provided in Table 6, highlighting their focus areas, key innovations, and their primary architectural paradigm.

These studies collectively mark the progression from explicit, algorithmic deliberation to emergent, stochastic intelligence. They serve as reference points for the fundamental differences in how adaptability, coordination, and strategic reasoning are implemented in each paradigm, illustrating the conceptual divide captured by our framework.

Table 6 Foundational studies and breakthroughs in Agentic AI by paradigm

Study	Focus area	Key contribution	Primary paradigm
Planning and decision-making (Kaelbling et al. 1998)	Theoretical foundations	Formalized the MDP/POMDP models for algorithmic decision-making under uncertainty	Symbolic
The SOAR cognitive architecture (Laird 2022)	Cognitive models	Provided a unified theory of cognition and a general cognitive architecture for symbolic agents	Symbolic
Strategic reasoning with language models (Gandhi et al. 2023)	Strategic reasoning	Demonstrated emergent long-term planning and belief modeling in multi-agent settings using LLMs	Neural
AI alignment: a comprehensive (Survey Ji et al. 2023)	Alignment architecture	Proposed scalable frameworks for robustness and interpretability, primarily focused on neural systems	Neural
Human–AI interaction on TikTok (Kang and Lou 2022)	Human–AI interaction	Explored agency trade-offs and context-driven decisions in dynamic social platforms powered by algorithms	Neural
Orchestration logics for AI platforms (Weber et al. 2024)	System integration	Analyzed orchestration strategies for agentic workflows, a key concern for neural pipeline management	Neural

4.3 Architectural paradigms: a mechanistic comparison

The advent of large language models (LLMs) has solidified the neural/generative paradigm, which operates on principles fundamentally incompatible with its symbolic predecessor. Modern agentic frameworks leverage LLMs as generative engines within software pipelines, explicitly departing from classical cognitive loops. Their core innovation lies in dynamic context management, prompt engineering, and tool composition. Table 7 provides a comparative analysis based on their primary orchestration mechanism, categorizing them under the neural paradigm and avoiding conceptual retrofitting into symbolic models.

This analysis underscores that these frameworks form the backbone of the neural paradigm, designed for practical task completion through orchestration, not for simulating internal cognitive processes. Mapping them to PPAR or BDI obscures their true innovative mechanics, which are defined by prompt-driven stochasticity, not algorithmic symbol manipulation.

4.4 Domain-specific implementations: a paradigm-driven analysis

Agentic AI frameworks are being deployed across sectors where autonomy and adaptability are essential. The choice of paradigm is critically influenced by domain-specific con-

Table 7 Orchestration mechanisms of modern neural agentic frameworks

Framework	Primary mechanism	Functional paradigm and representative applications
LangChain (Mavroudis 2024)	Prompt chaining	Orchestrates linear sequences of LLM calls and API tools. Replaces symbolic planning with stochastic generation of next steps. Applications: multi-step workflow automations, automated medical reporting (Huh et al. 2023)
AutoGen (Wu et al. 2023)	Multi-agent conversation	Facilitates structured dialogues between collaborative LLM agents. Replaces monolithic control with emergent problem-solving through conversation. Applications: collaborative task solving, economic research coordination (Dawid et al. 2025)
CrewAI (Venkatesh et al. 2024)	Role-based workflow	Assigns roles and goals to a team of agents, managing their interaction workflow. Replaces centralized scheduling with dynamic, role-driven process management. Applications: market analysis and risk modeling (Chandrashekar et al. 2025)
Semantic kernel (Kothapalli 2024)	Plugin/function composition	Connects LLMs to pre-written code functions ("skills"). Replaces integrated actuation with stochastic planning of plugin sequences. Applications: breaking down high-level user intents into executable skills
LlamaIndex (Gheorghiu 2024)	Retrieval-augmented generation (RAG)	Provides sophisticated data connectors and indexing. Replaces internal symbolic knowledge bases with on-demand, external context retrieval. Applications: financial sentiment analysis (Konstantinidis et al. 2024), enhancing information retrieval for research (Komineni et al. 2024)

straints—ethical, regulatory, or epistemic. The following implementations exemplify how each paradigm is applied.

Domain-specific applications and paradigm choices

The application of Agentic AI reveals a distinct paradigm split influenced by the core requirements of each sector. In *healthcare*, where *safety and compliance are paramount*, applications diverge clearly along architectural lines. Symbolic systems, such as rule-based clinical decision support tools, are predominantly employed for predictable and auditable tasks. In contrast, the flexibility of neural paradigms is leveraged for tasks like generating structured medical reports (Huh et al. 2023) and powering on-premise edge agents (Basil et al. 2024); however, these neural frameworks are often contained within deterministic tool-chaining pipelines to ensure the reliability required in clinical settings.

This pattern of complementary paradigm use is also evident in *finance, a domain demanding high accuracy and auditability*. Here, neural frameworks dominate tasks involving complex data synthesis and analysis. For instance, CrewAI's role-based workflow is applied to market analysis (Chandrashekar et al. 2025) as it provides a clear, auditable trail of agent actions. Similarly, LlamaIndex-powered models for financial sentiment (Konstantinidis et al. 2024) demonstrate how neural systems use Retrieval-Augmented Generation (RAG) to ground their stochastic outputs in verified data, thereby reducing hallucination. Despite this, symbolic systems maintain a critical role in high-frequency trading and core regulatory logic where absolute determinism is non-negotiable.

Finally, in *scientific research, which requires profound epistemic rigor*, the choice of paradigm is dictated by the nature of the intellectual task. The deployment of AutoGen to coordinate multi-agent conversations for economic research (Dawid et al. 2025) exemplifies the neural paradigm's strength in simulating collaborative, exploratory discovery and critique. This stands in direct contrast to the role of symbolic systems, which remain the bedrock for theorem proving and logical inference, highlighting a fundamental architectural choice between exploratory generation and deductive reasoning.

These implementations demonstrate that the paradigm choice is not merely technical but is decisively shaped by domain-specific needs, validating the need for a clear taxonomic framework to classify and select appropriate architectures.

4.5 Coordination protocols: from algorithmic contracts to emergent conversation

A critical yet often underexplored aspect of multi-agent systems (MAS) is the fundamental distinction in their coordination mechanisms. A deeper examination reveals that these strategies are a primary differentiator between the two paradigms, reflecting their core architectural principles: *explicit algorithms* in the symbolic paradigm versus *emergent, stochastically-guided behavior* in the neural paradigm.

Within the *symbolic paradigm*, coordination is achieved through pre-defined, algorithmic protocols rooted in decades of distributed AI research. These protocols are engineered to ensure predictable, verifiable, and fault-tolerant interactions, making them indispensable for critical systems where correctness is paramount. A quintessential example is the *contract net protocol (CNP)* (Xu and Weigand 2001), a classic negotiation framework where a manager agent announces a task through a "call for proposals." Other agents then evaluate their capabilities and submit bids, leading the manager to award the contract to the most suitable agent. This process, analogous to an auction, is extensively applied in domains like manufacturing and logistics scheduling. Another foundational strategy is the *blackboard system* (Craig 1988), where a shared memory space acts as a central coordination point. Specialist agents, akin to experts surrounding a physical blackboard, monitor this space for relevant data and contribute their expertise incrementally to build towards a solution. This approach is highly effective for complex, unstructured problems like medical diagnosis or signal interpretation. Furthermore, *market-based approaches* facilitate coordination through a virtual economy where agents buy and sell services or resources, providing a decentralized method for resource allocation in networked systems.

In direct opposition, coordination within the *Neural Paradigm* is not typically governed by hard-coded protocols. Instead, it emerges as a property of **structured conversation and prompt-driven orchestration** (Borghoff et al. 2025; Wang et al. 2025; Brodimas et al.

2025). Here, a central orchestrator (often an LLM itself) or the agents themselves leverage their generative capabilities to dynamically assign roles, manage dialogue, and synthesize results. This can manifest in several distinct patterns. *Conversation-based coordination* (Casella and Wang 2025; Luo et al. 2025; Tran et al. 2025), exemplified by frameworks like AutoGen, achieves collaboration through structured conversational loops where agents with defined roles interact within a group chat, with the LLM's context window managing the interaction state. A more explicit variant is the *role-based workflow* (Berti et al. 2024) (e.g., CrewAI), where a higher-level orchestrator assigns tasks based on pre-defined roles and goals, though the routing decisions are still driven by LLM-based reasoning rather than deterministic algorithms. Lastly, *dynamic context management* (Cheung et al. 2025; Wang and Duan 2024) (e.g., LangGraph) implements coordination through state machines that control information flow between nodes; the graph structure defines possible paths, but the specific execution is determined stochastically by the LLM's output at each step.

The fundamental dichotomy between these coordination strategies is summarized in Table 8, which highlights the core operational differences.

This analysis confirms that the paradigm shift extends to the very fabric of multi-agent coordination. The symbolic paradigm offers *verifiable reliability* through rigorously engineered protocols, while the neural paradigm offers *adaptable emergence* through learned conversation patterns. This critical distinction is essential for understanding the capabilities, risks, and appropriate applications of modern MAS, thereby further validating the necessity of the dual-paradigm framework presented in this survey.

Table 8 A dual-paradigm comparison of multi-agent coordination mechanisms

Feature	Symbolic/classical paradigm	Neural/generative paradigm
Primary mechanism	Algorithmic protocols (e.g., contract net, blackboard)	Structured conversation and prompt orchestration
State management	Explicit, often centralized (e.g., Manager in CNP, Blackboard)	Implicit, managed within the LLM's context window
Decision process	Deterministic or probabilistic based on explicit rules	Stochastic generation of next action/response
Flexibility	Low; protocols are fixed and designed for anticipated scenarios	High; can adapt to novel coordination patterns not explicitly programmed
Verifiability	High; the protocol's logic can be formally verified and audited	Low; the emergent coordination path is opaque and difficult to trace
Key frameworks	JADE, JaCaMo, early SOAR systems	AutoGen, CrewAI, LangGraph
Example	A manager agent uses CNP to auction a delivery task to the lowest-bidding drone agent	An orchestrator LLM manages a conversation between a programmer agent, a tester agent, and a writer agent to collaboratively build software

4.6 Evaluating agency: beyond accuracy

The evaluation of Agentic AI systems presents a fundamental challenge that distinguishes it from the assessment of traditional AI models. Simple metrics like accuracy are wholly insufficient. Measuring “agency” requires quantifying a system’s capacity for sustained, goal-directed behavior in dynamic environments, necessitating a multi-dimensional evaluation framework that accounts for paradigm-specific mechanisms of action.

The core challenge lies in the fact that agency is not a monolithic property but a spectrum encompassing *autonomy*, *task success*, *efficiency*, and *robustness*. Consequently, evaluation must be tailored to the architectural paradigm.

In the *symbolic paradigm*, evaluation has historically focused on *verifiability*. Key metrics include *goal completion fidelity*, which measures the percentage of pre-defined sub-goals correctly achieved in a plan, and *plan optimality*, which compares the cost (e.g., time, steps) of an agent’s generated plan against a known optimal solution. Furthermore, assessment involves verifying *logical soundness* through formal methods to ensure rule sets cannot derive contradictory or unsafe actions, and rigorously testing *Edge Case Handling* against rare but critical scenarios either explicitly encoded in or missing from the agent’s knowledge base.

Conversely, in the *neural paradigm*, evaluation is inherently more complex due to inherent stochasticity. While benchmarks like AgentBench (Liu et al. 2023) and GAIA (Mialon et al. 2023) represent a shift towards holistic assessment, they have limitations. Metrics must be designed to capture emergent capabilities and failures. This includes evaluating *long-horizon task success* on complex, multi-step tasks (e.g., “research a topic and write a report with citations”), often measured by final outcome quality as judged by humans or a powerful LLM “judge.” Other critical dimensions are *context window and memory management*, which assess an agent’s ability to utilize information across extended interactions; *tool use proficiency*, encompassing tool selection accuracy, call sequence efficiency, and error recovery; *robustness to prompts*, testing consistency across instruction rephrasings and resilience to injection attacks; and practical *cost and latency* metrics, measuring computational expense (e.g., total tokens, API calls) and time-to-completion, which are crucial for real-world deployment.

A comprehensive evaluation framework for Agentic AI must therefore integrate these dimensions. It is not enough for an agent to eventually succeed at a task; it must do so efficiently, reliably, and in a manner that is transparent and auditable where required. This typically involves a synergistic combination of automated metrics (e.g., success rate, number of steps), human evaluation for qualitative judgment of output coherence and usefulness, and adversarial testing (e.g., “red teaming”) to probe for specific failure modes like hallucination or goal divergence.

This paradigm-aware approach to evaluation—where symbolic systems are judged on verifiability and neural systems on robust adaptability—is essential for the responsible development and deployment of autonomous agents. It moves the field beyond simple benchmarks towards a more nuanced understanding of what it means for an AI system to be truly “agentic.”

5 Discussion: synthesis, implications, and future directions

Our paradigm-aware analysis yields several pivotal insights that chart the current and future state of Agentic AI. The bifurcation of research across symbolic and neural paradigms reveals that effective governance, evaluation, and advancement cannot follow a one-size-fits-all approach but must be strategically tailored to each architectural lineage.

5.1 Key insights from the paradigm-aware analysis

Our analysis reveals several fundamental insights:

5.1.1 Paradigm-market fit

The review demonstrates a clear *paradigm-market fit*, wherein symbolic and hybrid architectures dominate safety-critical applications like healthcare and robotics, while pure neural systems thrive in data-rich, adaptive domains such as finance and education. This finding validates our core thesis that paradigm selection is a strategic decision driven by domain constraints rather than technological superiority.

5.1.2 Governance imbalance

The taxonomy exposes a significant *governance imbalance*; while ethical challenges within the neural paradigm are the subject of intense research, the governance of modern, complex symbolic systems remains a critically underexplored area. This creates a vulnerability in safety-critical systems where symbolic AI is predominantly deployed.

5.1.3 Evaluation challenge

The successful classification of all 90 studies by this dualist framework validates its comprehensive coverage and utility as a robust tool for literature analysis and future research design. However, it also highlights the *evaluation challenge* of creating paradigm-specific benchmarks that can accurately measure the distinct capabilities and failure modes of each architectural approach.

5.2 Research gaps: a paradigm-specific roadmap

The development of Agentic AI is constrained by significant, unresolved challenges. However, these research imperatives differ profoundly between paradigms, with a particularly pressing need for work on hybrid architectures. A clear roadmap is required to address these paradigm-specific gaps and guide future research investment. The challenges in core areas like evaluation, reasoning, safety, and governance are not monolithic; they manifest in uniquely critical ways depending on whether a system is built on a symbolic or neural foundation.

Table 9 provides a detailed breakdown of these gaps and their corresponding research imperatives across nine critical areas. It highlights that while symbolic paradigms struggle with brittleness and scalability in open-world environments, neural paradigms are plagued

Table 9 Paradigm-specific research gaps and imperatives in agentic AI

Gap area	Symbolic paradigm challenges	Neural paradigm challenges	Research imperatives
Evaluation and benchmarks (Moshkovich et al. 2025; Zhuge et al. 2024)	Lack of standardized metrics for scalability and robustness of logical reasoning in complex, open-world environments	Current benchmarks (e.g., <i>AgentBench</i> Liu et al. 2023), <i>GAEA</i> (Mialon et al. 2023)) fail to adequately test for subtle misalignments, prompt robustness, and the true cost of context management	Develop paradigm-specific benchmarks. <i>Symbolic</i> Test logical soundness and failure predictability. <i>Neural</i> Test for hallucination under pressure, prompt injection resilience, and multi-session consistency
Reasoning and adaptability (Wu et al. 2025; Zhuang et al. 2025)	Systems are brittle; they fail catastrophically when faced with novel scenarios or exceptions not covered by their rules	Agents struggle with true, abstract reasoning and value-laden judgment. Their "reasoning" is often just sophisticated pattern matching that can break down	<i>Hybrid research</i> Investigate neuro-symbolic architectures where neural components handle pattern recognition and symbolic modules enforce rigorous reasoning and constraint checking
Long-term autonomy and memory	Can maintain a persistent, symbolic state but struggle to learn and update their world model from experience in a scalable way	Context window limitations create agents with severe amnesia across sessions. Statelessness prevents cumulative learning and building long-term relationships	<i>Symbolic</i> Research on efficient belief revision. <i>Neural</i> Develop architectures for external, structured memory that agents can reliably read from and write to
AI Infrastructure dependence R(adanliev 2025)	Performance is often constrained by the scalability of theorem provers and logic engines, which are sensitive to hardware architecture. Less dependent on massive cloud clusters but requires specialized, reliable compute	Extreme dependence on vast, expensive cloud compute for training and inference. Creates environmental costs, centralizes power, and creates vulnerabilities to supply chain and geopolitical disruptions	Develop energy-efficient and decentralized computing paradigms. Research model distillation, sparse architectures, and hybrid cloud-edge deployment to reduce reliance on monolithic infrastructure
Human-AI interaction and interface design (Schmidt and Loidolt 2023)	Interfaces are typically explicit (e.g., config files, rule editors). The goal is to augment human intelligence with transparent, predictable tools. The distinction between user and agent is clear	The goal is often a collaborative, conversational partner. Risk of creating opaque "oracles" that users over-trust. Challenges in designing intuitive interfaces for steering, interrupting, and interpreting the stochastic outputs of neural agents	Establish principles for paradigm-aware HCI. <i>Symbolic</i> Develop advanced visualization for logic and state. <i>Neural</i> Research intuitive methods for context steering, confidence communication, and collaborative task management
Trust and transparency (Lakkamraju 2025; Borghoff et al. 2025)	"How" decisions are made is transparent (the logic trace), but "why" a specific rule exists can be opaque	Both "how" and "why" are opaque. Explanations are post-hoc and often unreliable. This is the primary barrier to high-stakes deployment	<i>Symbolic</i> : Research on making goal structures and utility functions explicable. <i>Neural</i> Fundamental research on mechanistic interpretability and generating faithful, real-time explanations
Safety and alignment (Hellrigel-Holderbaum and Dung 2025; Zhang et al. 2025)	Risk of "perverse instantiation"—perfectly executing a flawed or oversimplified goal specification with catastrophic results	Vulnerability to prompt injection, goal drift, and value misgeneralization. Aligning a stochastic model to complex human values is an unsolved problem	Paradigm-specific strategies <i>Symbolic</i> Formal verification of goals and constraints. <i>Neural</i> Advanced red teaming, adversarial training, and "constitutional" oversight mechanisms

Table 9 (continued)

Gap area	Symbolic paradigm challenges	Neural paradigm challenges	Research imperatives
Interoperability and integration (Jeong 2025; Li and Xie 2025)	Difficult to integrate with the messy, unstructured data of the real world and modern software ecosystems	Excel at using tools via APIs but struggle with true, semantic understanding of what a tool does, leading to misuse	Develop standards and middle-ware for <i>paradigm bridging</i> . Create APIs that allow neural agents to query symbolic reasoners for validation and symbolic systems to leverage neural networks for perception
Governance and accountability (Tenant et al. 2023; Rossi and Mattei 2019)	Liability is more straightforward (flawed logic can be traced) but frameworks for auditing complex rule sets are needed	A profound "attribution gap" exists. Legal frameworks are unprepared for harm caused by emergent, stochastic behavior	Urgently develop paradigm-specific regulatory models. <i>Symbolic</i> Audit trails for decision logic. <i>Neural</i> : Mandatory context logging, output watermarking, and potentially new forms of developer liability

by opaqueness, an inability to perform verifiable reasoning, and a massive dependence on unsustainable compute infrastructure. The path forward necessitates a dual strategy: strengthening each paradigm within its core competencies while aggressively pursuing neuro-symbolic integration to create agents that are both adaptable and trustworthy. This table serves as a framework for targeting research efforts where they are most needed to overcome the current limitations of Agentic AI.

5.3 Future directions: the path to hybrid intelligence

The most active and promising research in emerging technologies explicitly seeks to integrate both paradigms, confirming that the most viable *path forward is hybrid*. This strategic direction leverages the complementary strengths of symbolic reliability and neural adaptability rather than pursuing the dominance of either paradigm alone.

Agentic AI systems are rapidly evolving beyond static task automation into dynamic, collaborative, and adaptive entities (Ma et al. 2025). Their future development will hinge on interdisciplinary advances, technological convergence, and—critically—a paradigm-aware approach to design that seeks to integrate the strengths of both symbolic and neural lineages into robust hybrid architectures.

A summary of these paradigm-aware trajectories is presented in Table 10, which outlines the specific research and integration priorities for each paradigm's evolution.

5.3.1 Analysis of strategic trajectories

The bifurcated future outlined in Table 10 leads to one overriding conclusion: the paramount direction is *architectural integration*. The goal is to forge a new class of hybrid systems that leverage the reliability of symbolic reasoning and the adaptability of neural generation.

- Neuro-symbolic integration as the keystone** The most profound progress will come from research that successfully couples neural networks for perception and pattern recognition with symbolic engines for reasoning and constraint checking. This is the most promising path to overcoming the brittleness of pure symbolism and the opacity of pure

Table 10 Paradigm-aware strategic trajectories for Agentic AI

Strategic direction	Symbolic paradigm evolution	Neural paradigm evolution
Multi-agent ecosystems	Defining verifiable communication protocols and interaction contracts for hybrid agent teams	Specializing in emergent, role-based collaboration and negotiation (Huang et al. 2025; Bachrach et al. 2020) (e.g., CrewAI, AutoGen, LangGraph)
Technological convergence	Providing the reliable, verifiable logic layer for cyber-physical systems and smart infrastructure	Acting as the adaptive interface for integrating with Internet of Things (IoT), robotics, blockchain, and quantum computing (Sultanow et al. 2025; Radanliev 2025)
Self-evolving architectures	Research into automated theorem proving and logical rule discovery for system self-improvement	Advancing meta-learning and feedback-driven optimization (Ma et al. 2025) for architecture tuning and deployment-aware adaptation
Human-AI collaboration	Enabling interfaces where humans can directly inspect, debug, and modify an agent's logical rule set and goals	Creating intuitive interfaces for shared intent and cognitive/emotional responsiveness Schmidt and Loidolt (2023) via natural language
Governance-first design	Formal verification of goal structures and safety constraints for embeddable governance modules	Developing techniques for embedded ethics, policy enforcement, and global accountability Gabison and Xian (2025) within stochastic systems (e.g., IBM Governance Stack)
Scientific discovery	Encoding scientific laws and methodological rigor for agent-led hypothesis generation	Driving agent-led inquiry and results analysis (Koutra et al. 2025; Lu et al. 2024) in platforms like Sakana AI Scientist (Lu et al. 2024) and microsoft discovery
Research priorities	Establishing benchmarks for logical soundness, verifiability, and interoperability standards	Establishing metrics for moral alignment, cognitive modeling, and alignment (Cervantes et al. 2020; AgentBench Liu et al. 2023)

neural approaches.

- *Paradigm-specialized roles in ecosystems* Future multi-agent ecosystems (Huang et al. 2025; Bachrach et al. 2020) will not be homogenous. They will consist of specialized agents—some highly neural for creative tasks, some highly symbolic for regulatory compliance—that communicate through standardized protocols. The orchestration of

such hybrid swarms is a critical research frontier.

- *A dual-track approach to governance* The development of safety and governance mechanisms (Gabison and Xian 2025) must continue on two tracks: advancing formal methods for symbolic verifiability and developing new statistical, training-based methods for neural alignment. The ultimate governance framework for a hybrid agent will need to seamlessly combine both.
- *Convergence as amplification* The integration with other technologies (Sultanow et al. 2025; Radanliev 2025) will amplify the capabilities of both paradigms. Neural agents will manage real-time sensor data from IoT, while symbolic modules will ensure the decisions made from that data are safe and compliant.

5.4 Ethical and governance challenges: a paradigm-specific analysis

As Agentic AI systems gain autonomy and are deployed in critical domains, they introduce a complex spectrum of ethical and governance concerns (Raza et al. 2025; Gabison and Xian 2025; Ranjan et al. 2025). A critical oversight in current discourse is the treatment of these challenges as monolithic. The risks and requisite mitigation strategies differ profoundly between the symbolic and neural paradigms, demanding a paradigm-aware approach to oversight.

A synthesis of these issues is presented in Table 11, which outlines the core challenges and their paradigm-specific manifestations and governance implications.

5.4.1 Analysis and summary

The bifurcation of ethical challenges detailed in Table 11 leads to several critical conclusions. First, it becomes evident that *effective governance cannot be architecturally agnostic*. Regulation and ethical oversight must be predicated on the underlying paradigm; a requirement for "full explainability," for instance, is feasible for a symbolic system but may be technologically impossible for a pure neural agent, thus necessitating the development of alternative compliance mechanisms.

Furthermore, the rise of *hybrid systems compounds ethical complexity*. An agent that blends paradigms inherently inherits the governance challenges of both. A neuro-symbolic architecture, for example, requires a framework capable of auditing its deterministic symbolic logic while simultaneously monitoring its neural components for stochastic failures, creating a significantly more demanding oversight burden.

Conversely, *the attribution gap presents a specific crisis for the neural paradigm*. The fundamental question of "Who is liable?" is most acute here, as its diffuse and stochastic nature directly challenges legal frameworks built on principles of direct causation and intent. This may ultimately require the establishment of new forms of strict liability for developers and operators.

Finally, these distinctions mean that *effective human-AI collaboration is inherently paradigm-dependent*. Designing appropriate human oversight requires a deep understanding of the agent's core mechanics. The process of overseeing a symbolic agent is analogous to supervising a junior programmer—it involves checking their logical steps. In stark contrast, overseeing a neural agent is more akin to supervising a brilliant but unpredictable intern—it requires carefully steering their context and interpreting their often-opaque outputs.

Table 11 Paradigm-specific ethical and governance challenges in Agentic AI

Challenge	Symbolic paradigm manifestation	Neural paradigm manifestation	Governance and mitigation strategies
Accountability and liability (Chan et al. 2023; Tóth et al. 2022)	Failure due to flawed logic or unhandled edge cases. Liability is potentially traceable to programmers or system designers	Failure due to stochastic outputs, prompt injection, or training data biases. Liability is diffuse and difficult to attribute	Paradigm-specific standards: <i>Symbolic</i> Code verification, formal proof of correctness. <i>Neural</i> Output watermarking, robust prompt shielding, audit trails for context history
Transparency and explainability (Baron 2025; Chan et al. 2024)	Inherently high. Reasoning trace is a sequence of logical steps or rule firings. "Why?" is answerable	Inherently low. "Reasoning" is an emergent property of model activations. "How?" is often unanswerable; "Why?" is inferred	<i>Symbolic</i> Leverage native explainability. <i>Neural</i> Invest in SHAP/LIME-style post-hoc explanations and mandatory decision logs. <i>Hybrid</i> Use symbolic modules to generate explanations for neural decisions
Bias and Fairness (Singh and Ngu 2025; Yadav 2024)	Bias arises from explicit, hand-coded rules or knowledge bases. Easier to identify but hard to root out if foundational	Bias is latent in training data and amplified stochastically. Pervasive and subtle, emerging in novel contexts	<i>Symbolic</i> Rigorous logic audits, diverse design teams. <i>Neural</i> Continuous bias monitoring, curated fine-tuning datasets, adversarial debiasing
Safety and misalignment (Hellrigel-Holderbaum and Dung 2025; Zhang et al. 2025)	Risk of "perverse instantiation" where agents exploit literal, rigid goals with unintended consequences	Risk of goal drift, prompt hacking, and value misgeneralization where agents pursue correlated but incorrect proxies	<i>Symbolic</i> Comprehensive failure mode testing. <i>Neural</i> Red teaming, constitutional AI, and harmlessness training. <i>Universal</i> Sandboxed testing environments
Autonomy vs. control (Feng et al. 2025; Talam 2025)	Human oversight is typically designed as explicit veto points or permission gates within a deterministic loop	Human oversight is fuzzy, often implemented as "human-in-the-loop" feedback, which can be ignored or gamed by the agent	Define "meaningful human control" by paradigm. <i>Symbolic</i> Clear interrupt signals. <i>Neural</i> Confidence thresholding for automatic deferral and nuanced steering mechanisms
Security and resilience (Narajala and Narayan 2025; Khan et al. 2024)	Vulnerabilities include logic bombs, sensor spoofing, and exploiting algorithmic flaws	Vulnerabilities include prompt injection, training data poisoning, and adversarial attacks on embeddings	Paradigm-specific defense: <i>Symbolic</i> Formal verification, intrusion detection. <i>Neural</i> Advanced prompt hardening, detection of out-of-distribution inputs, data provenance

5.5 Limitations and research constraints

While this review provides a comprehensive synthesis of Agentic AI research, several limitations must be acknowledged that also point to broader constraints in the field:

- *Temporal dynamics* The rapid evolution of the neural paradigm means that some very recent developments may not be fully captured, despite our search extending to early 2025.
- *Transparency constraints* Many state-of-the-art neural agentic systems are proprietary, limiting access to architectural details and performance metrics.
- *Methodological heterogeneity* The varied evaluation metrics across studies limited direct cross-paradigm benchmarking.
- *Classification challenges* Assigning hybrid architectures to a single paradigm involved

necessary simplification in some cases.

These limitations highlight the challenges of conducting systematic reviews in a nascent and fast-paced field with multiple co-existing paradigms.

6 Conclusion

Agentic AI represents a fundamental paradigm shift in the design of intelligent systems, but its rapid evolution has led to a fragmented and often anachronistic understanding of the field. This review has addressed this confusion by introducing and validating a novel conceptual framework: the existence of two distinct lineages of Agentic AI—the *symbolic/classical* and the *neural/generative*—each with fundamentally different operational mechanics, strengths, and limitations.

Our analysis demonstrates that the common practice of *conceptual retrofitting*—describing modern LLM-orchestrated agents with the language of symbolic systems (e.g., PPAR loops, BDI)—obscures their true nature and impedes progress. Through a systematic, paradigm-aware review of the literature, we have established that the architectural divide is both real and meaningful. Symbolic systems excel in environments requiring safety, verifiability, and explicit logic, while neural systems thrive in domains requiring adaptability, pattern recognition, and operation on unstructured data.

The most significant contribution of this work is the demonstration that effective governance, evaluation, and advancement of Agentic AI cannot follow a one-size-fits-all approach but must be strategically tailored to each architectural lineage. This paradigm-specific approach reveals that the most productive path forward is hybrid, not isolated. The future of Agentic AI lies in the strategic integration of neuro-symbolic architectures that leverage the complementary strengths of symbolic reliability and neural adaptability.

This dual-paradigm framework provides the essential analytical lens to move the field beyond a simple catalog of technologies toward a coherent theory of architectural design in Agentic AI. It offers researchers, engineers, and policymakers a precise vocabulary and a functional taxonomy to classify systems, evaluate their capabilities and risks appropriately, and make informed design choices.

Ultimately, the development of Agentic AI is not merely a technical challenge—it is a sociotechnical one. Its success will depend on whether we can architect systems that are not only powerful but also trustworthy. This requires a conscious and deliberate effort to build hybrid intelligence—systems that are both adaptable and reliable, both creative and sound. By recognizing and embracing the distinct nature of these two architectural lineages, we can steer this transformative technology toward a future where agentic systems truly serve as trusted collaborators in scientific discovery, in providing fair and accessible services, and in forming the robust, verifiable backbone of critical infrastructure.

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